Autocorrelation of a Rectangularly Windowed Linear Chirp

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Let $B > 0$ be the swept bandwidth and let $T_p > 0$ be the pulse duration. Using $j^2 = -1$, define the chirp

$$f(t) = \exp\left[j\left(\frac{\pi B}{T_p}t^2 - \pi B t\right)\right](u(t) - u(t - T_p)), \tag{1}$$

where $u(t)$ is the unit-step function. Thus $f(t)$ is supported on $[0, T_p]$. The values assigned to the step functions at the endpoints do not affect any integral below.

We use the autocorrelation convention

$$R_f(\tau) = \int_{-\infty}^{\infty} f(t) f^*(t - \tau)\, dt \tag{2}$$

Sinc conventions

Two conventions for the sinc function are common. This paper uses the unnormalized convention

$$\operatorname{sinc}(x) = \begin{cases} \dfrac{\sin x}{x}, & x \neq 0, \\ 1, & x = 0. \end{cases} \tag{3}$$

The alternative normalized convention is

$$\operatorname{sinc}_{\mathrm{N}}(x) = \begin{cases} \dfrac{\sin(\pi x)}{\pi x}, & x \neq 0, \\ 1, & x = 0, \end{cases} \qquad \operatorname{sinc}_{\mathrm{N}}(x) = \operatorname{sinc}(\pi x). \tag{4}$$

The unnormalized convention (3) is used throughout the remainder of the paper. Because B is measured in cycles per second, the corresponding dimensionless arguments contain the factor π, as in $\operatorname{sinc}(\pi B \tau)$.

The overlap interval

The first factor in the integrand is nonzero when $0 \le t \le T_p$. The second is nonzero when

$$\begin{aligned} 0 &\le t - \tau \le T_p, \\ \tau &\le t \le T_p + \tau. \end{aligned} \tag{5}$$

The product must be zero if

$$\min(T_p, T_p + \tau) \le t \quad \text{or} \quad t \le \max(0, \tau).$$

*I did the math. GPT wrote most of the paper based on my math.

If $|\tau| \geq T_p$, then

$$\begin{aligned} \min(T_p, T_p + \tau) \leq \max(0, \tau) \quad \text{so,} \\ R_f(\tau) = 0 \quad \text{if } |\tau| \geq T_p. \end{aligned} \tag{6}$$

Futhermore, the two factors overlap on

$$I_\tau = [\max(0, \tau),\, \min(T_p, T_p + \tau)]. \tag{7}$$

Now suppose $|\tau| \leq T_p$. The length and midpoint of the overlap interval are, respectively,

$$\begin{aligned} L(\tau) &= \min(T_p, T_p + \tau) - \max(0, \tau) = T_p - |\tau|, \\ c(\tau) &= \frac{\max(0, \tau) + \min(T_p, T_p + \tau)}{2} = \frac{T_p + \tau}{2}. \end{aligned} \tag{8}$$

Thus $I_\tau = [c(\tau) - L(\tau)/2, c(\tau) + L(\tau)/2]$.

Evaluation of the autocorrelation

Write the phase of the unwindowed chirp as

$$\phi(t) = \frac{\pi B}{T_p} t^2 - \pi B t. \tag{9}$$

On the overlap interval, the phase in the product $f(t)f^*(t-\tau)$ satisfies

$$\begin{aligned} \phi(t) - \phi(t-\tau) &= \frac{\pi B}{T_p}\left[t^2 - (t-\tau)^2\right] - \pi B\left[t - (t-\tau)\right] \\ &= \frac{\pi B}{T_p}(2t\tau - \tau^2) - \pi B\tau \\ &= \frac{2\pi B\tau}{T_p}\left(t - \frac{T_p + \tau}{2}\right) \\ &= \frac{2\pi B\tau}{T_p}(t - c(\tau)). \end{aligned} \tag{10}$$

Make the centered change of variable $x = t - c(\tau)$. By (8), its limits are $-L(\tau)/2$ and $L(\tau)/2$. For $0 < |\tau| \leq T_p$, the autocorrelation is therefore

$$\begin{aligned} R_f(\tau) &= \int_{I_\tau} \exp\big(j[\phi(t) - \phi(t-\tau)]\big)\, dt \\ &= \int_{-L(\tau)/2}^{L(\tau)/2} \exp\left(j\frac{2\pi B\tau}{T_p}x\right)\, dx \\ &= \frac{T_p}{j2\pi B\tau} \exp\left(j\frac{2\pi B\tau}{T_p}x\right)\Bigg|_{-L(\tau)/2}^{L(\tau)/2} \\ &= \frac{T_p}{\pi B\tau} \sin\left(\frac{\pi B\tau L(\tau)}{T_p}\right) \\ &= L(\tau)\ \mathrm{sinc}\left(\frac{\pi B\tau L(\tau)}{T_p}\right). \end{aligned} \tag{11}$$

At zero lag, direct evaluation gives

$$\begin{aligned} R_f(0) &= \int_{-\infty}^{\infty} |f(t)|^2 \, dt \\ &= \int_0^{T_p} 1 \, dt \\ &= T_p. \end{aligned} \tag{12}$$

This is also the continuous limit of (11), since $L(0) = T_p$ and $\operatorname{sinc}(0) = 1$.

Combining the overlap and non-overlap cases gives the exact result

$$\boxed{R_f(\tau) = \begin{cases} (T_p - |\tau|) \operatorname{sinc}\left(\dfrac{\pi B \tau (T_p - |\tau|)}{T_p}\right), & |\tau| \le T_p, \\ 0, & |\tau| > T_p. \end{cases}} \tag{13}$$

Equivalently, for $0 < |\tau| \le T_p$,

$$\begin{aligned} R_f(\tau) &= \frac{T_p}{\pi B \tau} \sin\left[\pi B \tau \left(1 - \frac{|\tau|}{T_p}\right)\right] \\ &= \frac{T_p}{\pi B |\tau|} \sin\left[\pi B |\tau| \left(1 - \frac{|\tau|}{T_p}\right)\right]. \end{aligned} \tag{14}$$

The second equality follows because the sine function is odd.

The sinc argument

$$\frac{\pi B \tau (T_p - |\tau|)}{T_p} = \pi B \tau \left(1 - \frac{|\tau|}{T_p}\right) \tag{15}$$

is piecewise quadratic in τ. Thus the result is a triangular overlap factor, $T_p - |\tau|$, multiplied by a sinc having a piecewise-quadratic argument. In particular, it is not merely a sinc by itself. The result is real and even, as is also clear from (13).

The parabola and the pulse endpoints

For positive lag, the numerator of the sinc argument contains

$$\begin{aligned} q(\tau) &= \tau(T_p - \tau) \\ &= -\left(\tau - \frac{T_p}{2}\right)^2 + \frac{T_p^2}{4}, \qquad 0 \le \tau \le T_p. \end{aligned} \tag{16}$$

This is a downward-opening parabola. It has its *maximum*, not its minimum, at $\tau = T_p/2$; its minima on the closed interval occur at $\tau = 0$ and $\tau = T_p$. In particular,

$$\begin{aligned} q'\left(\frac{T_p}{2}\right) &= 0, \\ q''(\tau) &= -2. \end{aligned} \tag{17}$$

It is important not to transfer the zero slope of q at its vertex to the endpoints of R_f. Set $\epsilon = T_p - \tau$ and approach T_p from inside the support. Since $\operatorname{sinc}(z) = 1 - z^2/6 + O(z^4)$ as $z \to 0$,

$$\begin{aligned} R_f(T_p - \epsilon) &= \epsilon \operatorname{sinc}\left[\pi B \epsilon \left(1 - \frac{\epsilon}{T_p}\right)\right] \\ &= \epsilon - \frac{\pi^2 B^2}{6}\epsilon^3 + O(\epsilon^4). \end{aligned} \tag{18}$$

Consequently, the interior one-sided derivatives are

$$R_f'(T_p^-) = -1, \qquad R_f'((-T_p)^+) = 1. \tag{19}$$

Outside the pulse-overlap interval, R_f is identically zero, so the exterior one-sided derivative at either endpoint is zero. Therefore R_f is not differentiable at $\tau = \pm T_p$, and its second derivative is not defined there. If only the interior expression is considered, its one-sided second derivative tends to zero at both endpoints. Thus a claim that the slope is zero and the second derivative is $-T_p$ at the endpoints does not hold for this autocorrelation.

Comparison with the large-time-bandwidth approximation

Figure 1 compares the exact result with the commonly used approximation $T_p \operatorname{sinc}(\pi B \tau)$ for $B = 10$ and $T_p = 2$. For these parameter values,

$$\begin{aligned} R_f\left(\frac{T_p}{2}\right) &= \frac{T_p}{2} \operatorname{sinc}\left(\frac{\pi B T_p}{4}\right) \\ &= \operatorname{sinc}(5\pi) \\ &= 0, \end{aligned} \tag{20}$$

so the requested marked point is $(T_p/2, R_f(T_p/2)) = (1, 0)$.

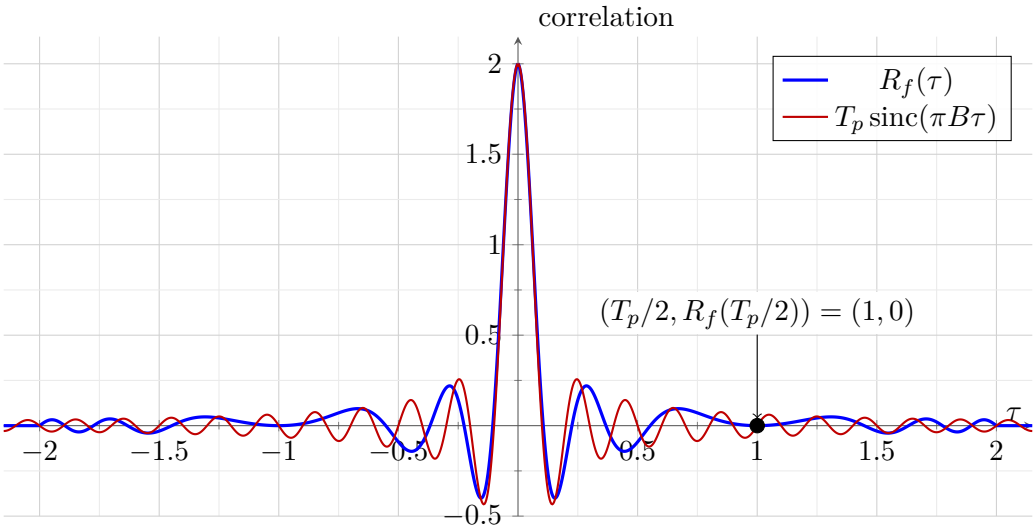


Figure 1: Exact chirp autocorrelation and the approximation $T_p \operatorname{sinc}(\pi B \tau)$ for $B = 10$ and $T_p = 2$.

There is a simple uniform error bound near the origin. For $0 \le \tau \le T_p$, let $L = T_p - \tau$ and $k = 2\pi B\tau/T_p$. The centered-integral form of the exact and approximate expressions gives

$$\begin{aligned} R_f(\tau) &= \int_{-L/2}^{L/2} e^{jkx}\,dx, \\ T_p\,\mathrm{sinc}(\pi B\tau) &= \int_{-T_p/2}^{T_p/2} e^{jkx}\,dx. \end{aligned} \tag{21}$$

Their difference consists only of the two omitted edge intervals. Direct integration, followed by the sine-difference identity, gives the exact error formula

$$\begin{aligned} |R_f(\tau) - T_p\,\mathrm{sinc}(\pi B\tau)| &= \frac{T_p}{\pi B\tau}\left|\sin(\pi B\tau) - \sin\left[\pi B\tau\left(1 - \frac{\tau}{T_p}\right)\right]\right| \\ &= \frac{2T_p}{\pi B\tau}\left|\cos\left[\pi B\tau\left(1 - \frac{\tau}{2T_p}\right)\right]\sin\left(\frac{\pi B\tau^2}{2T_p}\right)\right|. \end{aligned} \tag{22}$$

Using $|\sin z| \le |z|$ now gives the phase-sensitive pointwise bound

$$\boxed{|R_f(\tau) - T_p\,\mathrm{sinc}(\pi B\tau)| \le \tau\left|\cos\left[\pi B\tau\left(1 - \frac{\tau}{2T_p}\right)\right]\right|.} \tag{23}$$

This is never worse than the earlier bound by τ, but it is generally strictly better because it retains the cancellation between the two edge intervals. Notice also that a proposed bound proportional to $\tau T_p/10$ would have units of time squared, whereas the autocorrelation and its error have units of time; such a bound would need an additional inverse-time scale.

For $B = 10$ and $T_p = 2$, maximization of the right side of (23) on $0 \le \tau \le 0.2$ gives

$$\begin{aligned} \sup_{0\le\tau\le 0.2} \tau\left|\cos\left[10\pi\tau\left(1 - \frac{\tau}{4}\right)\right]\right| &= 0.2\cos\left(\frac{\pi}{10}\right) \\ &\approx 0.190211. \end{aligned} \tag{24}$$

The exact error formula is tighter still. In fact, its supremum on this interval occurs at $\tau = 0.2$ and is

$$\boxed{\begin{aligned} \sup_{0\le\tau\le 0.2} |R_f(\tau) - 2\,\mathrm{sinc}(10\pi\tau)| &= |R_f(0.2) - 2\,\mathrm{sinc}(2\pi)| \\ &= \frac{\sin(\pi/5)}{\pi} \\ &\approx 0.1870978568. \end{aligned}} \tag{25}$$

For completeness, this endpoint maximum can be verified without relying on a plot. Let $E_* = \sin(\pi/5)/\pi$. On $0 \le \tau \le E_*$, the elementary bound by τ gives an error no larger than E_*. On $E_* \le \tau \le 0.2$, the two positive factors

$$\frac{\sin(5\pi\tau^2/2)}{\tau} \quad \text{and} \quad \cos\left[\pi\left(10\tau - \frac{5}{2}\tau^2\right)\right] \tag{26}$$

are increasing. For the first factor this follows by differentiation and $\tan z < 2z$ for $0 < z \le \pi/10$; for the second it follows because its phase lies between $3\pi/2$ and 2π and increases with τ. Therefore the exact error is increasing on the remaining subinterval and attains its maximum at $\tau = 0.2$.

An approximation with cubic error

The approximation in the preceding section replaces both the exact overlap length and the exact sinc argument. A more accurate approximation retains the overlap length. For $0 \le \tau \le T_p$, define

$$\widetilde{R}_f(\tau) = (T_p - \tau)\,\mathrm{sinc}(\pi B \tau). \tag{27}$$

The factor π is required because B is measured in cycles per second and the paper uses the unnormalized sinc convention.

Let $L = T_p - \tau$. The exact expression and the approximation have the centered-integral representations

$$\begin{aligned} R_f(\tau) &= \int_{-L/2}^{L/2} \exp\!\left(j\frac{2\pi B\tau}{T_p}t\right)\,dt, \\ \widetilde{R}_f(\tau) &= \int_{-L/2}^{L/2} \exp\!\left(j\frac{2\pi B\tau}{L}t\right)\,dt. \end{aligned} \tag{28}$$

Set

$$\alpha = \frac{2\pi B\tau}{T_p}, \qquad \beta = \frac{2\pi B\tau}{L}. \tag{29}$$

For $0 \le \tau < T_p$, one has $0 \le \alpha \le \beta$. The identity for the difference of two cosines and the inequality $|\sin z| \le |z|$ give

$$\begin{aligned} |\cos(\alpha t) - \cos(\beta t)| &= 2\left|\sin\!\left(\frac{(\alpha+\beta)t}{2}\right)\sin\!\left(\frac{(\beta-\alpha)t}{2}\right)\right| \\ &\le \frac{\beta^2 - \alpha^2}{2}t^2. \end{aligned} \tag{30}$$

The imaginary parts in (28) vanish by symmetry. Consequently,

$$\begin{aligned} |R_f(\tau) - \widetilde{R}_f(\tau)| &\le 2\int_0^{L/2} |\cos(\alpha t) - \cos(\beta t)|\,dt \\ &\le (\beta^2 - \alpha^2)\frac{L^3}{24} \\ &= \frac{\pi^2 B^2 \tau^3}{6}\left(1 - \frac{\tau}{T_p}\right)\left(2 - \frac{\tau}{T_p}\right) \\ &\le \frac{\pi^2 B^2}{3}\tau^3. \end{aligned} \tag{31}$$

Thus

$$\boxed{|R_f(\tau) - (T_p - \tau)\,\mathrm{sinc}(\pi B\tau)| \le \frac{\pi^2 B^2 \tau^3}{6}\left(1 - \frac{\tau}{T_p}\right)\left(2 - \frac{\tau}{T_p}\right).} \tag{32}$$

The local expansion shows that the cubic order and its leading coefficient are exact:

$$\begin{aligned} R_f(\tau) - (T_p - \tau)\,\mathrm{sinc}(\pi B\tau) &= \frac{\pi^2 B^2}{3}\tau^3 - \frac{\pi^2 B^2}{2T_p}\tau^4 + O(\tau^5), \\ R_f(\tau) - (T_p - \tau)\,\mathrm{sinc}(\pi B\tau) &\sim \frac{\pi^2 B^2}{3}\tau^3 \qquad (\tau \to 0^+). \end{aligned} \tag{33}$$

Location of the first null

For a large time-bandwidth product BT_p, the first positive null near the origin is close to $1/B$. More precisely, the central null solves

$$B\tau\left(1-\frac{\tau}{T_p}\right)=1, \tag{34}$$

so, when $BT_p \geq 4$, it is

$$\tau_{\text{null}}=\frac{T_p}{2}\left(1-\sqrt{1-\frac{4}{BT_p}}\right)\approx\frac{1}{B}. \tag{35}$$